EART11200: Natural Scientist's Toolkit

Measuring Molecules







In this lecture we will:

- Introduce concepts, and this week's problem: ***measuring molecules***.
- Periodic table
  - We will not cover (ionic, covalent) bonding today, You will cover that in the Chemistry of the Earth part of this module. But information in appendix C2 if interested.

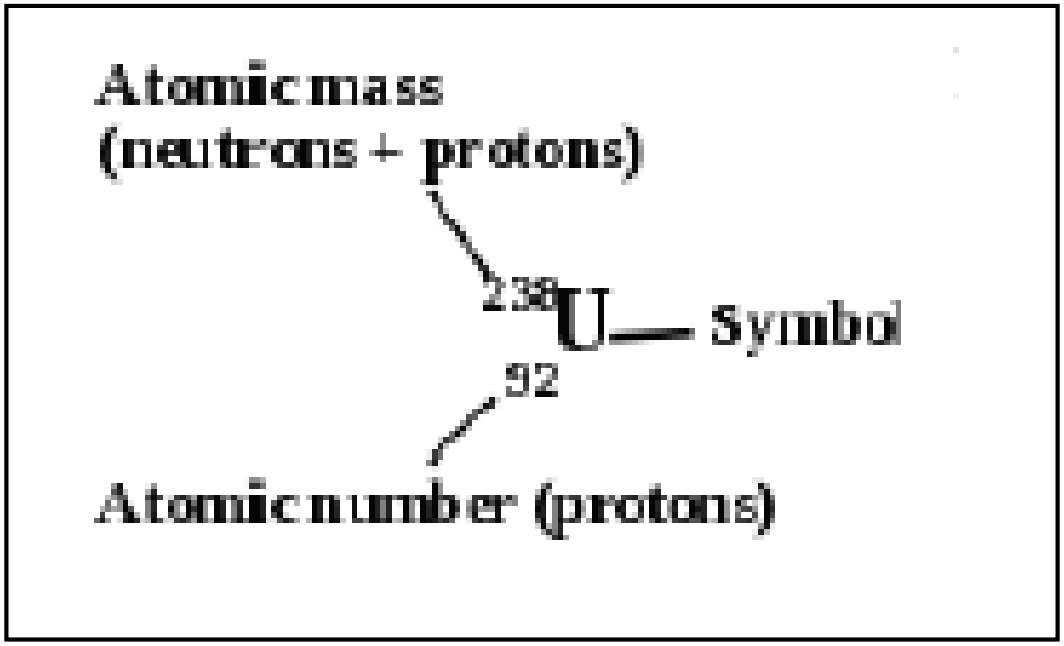
    eart11200_appendices
- Atoms, atomic mass, molecular mass, moles.
- Avogadro's number

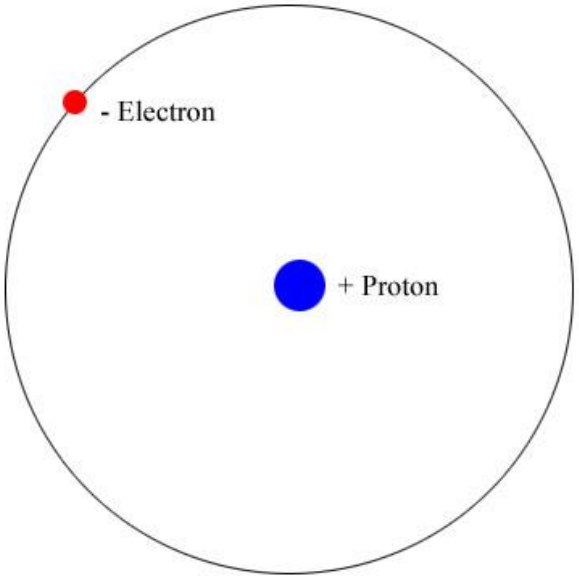
Atoms, hydrogen and Periodic Table

Atoms are composed of 3 basic particles: protons, neutrons and electrons:

Particle	Mass	Charge
Proton	1 atomic unit	+1
Neutron	1 atomic unit	0
Electron	negligible (0)	-1

Atoms are electrically neutral: therefore number protons = number electrons





The number of protons (=number of electrons) uniquely determines what sort of element that an atom is but (within limits) the number of neutrons can be different for different atoms of the same element—thus giving different *isotopes* of the element.

Size ~ 1E-10 m = 1 Å

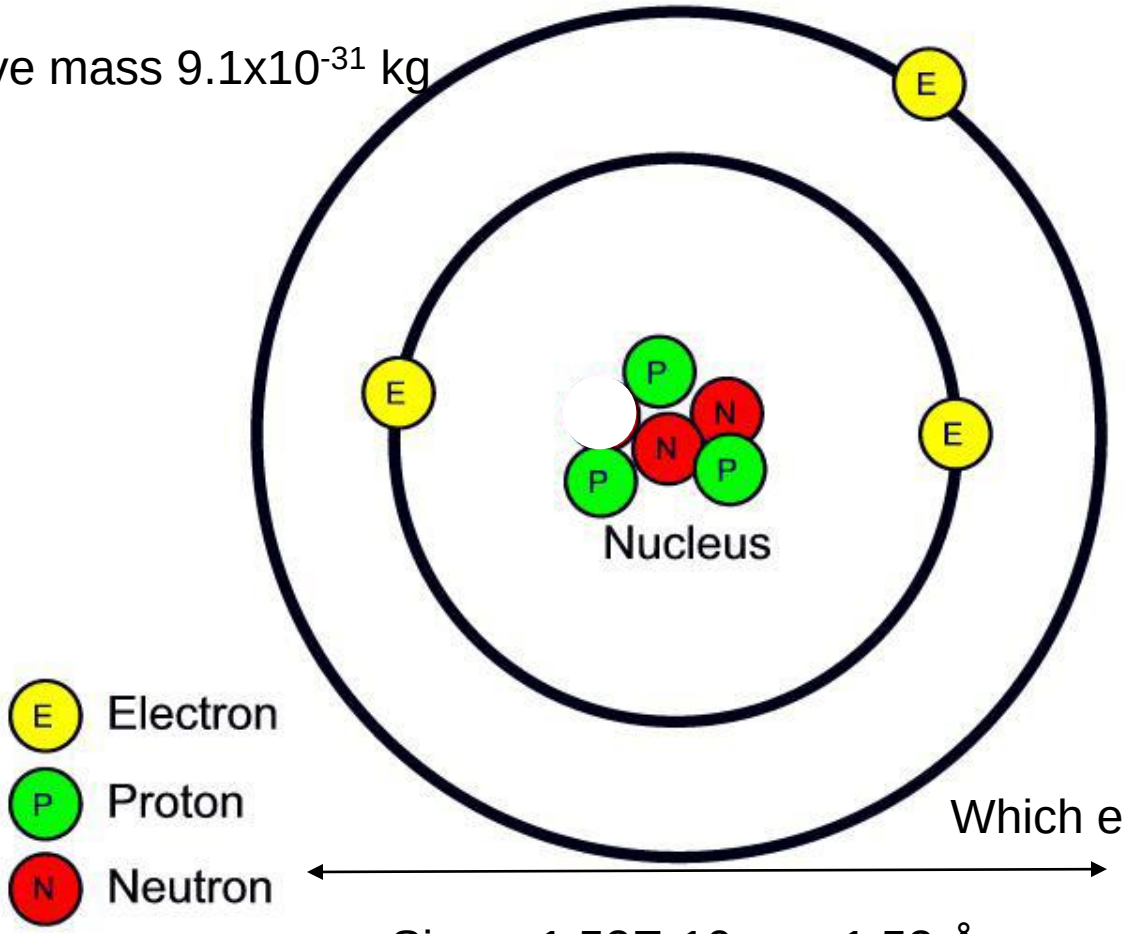
Shell model of atom

(neutrons help to bind the nucleus together)

Electrons hardly have any mass compared to protons and neutrons, which both have mass 1.66×10^{-27} kg

Electrons have mass 9.1×10^{-31} kg

Say we get rid of a neutron (e.g. fire high energy neutron at it to knock one out) – what happens? What element is it now?



Which element is this atom?

Size ~ 1.52E-10 m = 1.52 Å

Avogadro's number

The mass of a proton (1.66 E-24 g) is so small that we need another way to express quantities of atoms, for example a defined number of atoms we can measure in grams.
Take hydrogen, the simplest, smallest atom of just one proton and one electron. How many hydrogen atoms would weigh one gram?
(# of H atoms in 1 gram) x (mass of 1 H atom) = 1 gram
1/ (mass of 1 H atom) = (# of H atoms in 1 gram) = 6.02E23 = N_A
This is known as Avagadro's number.

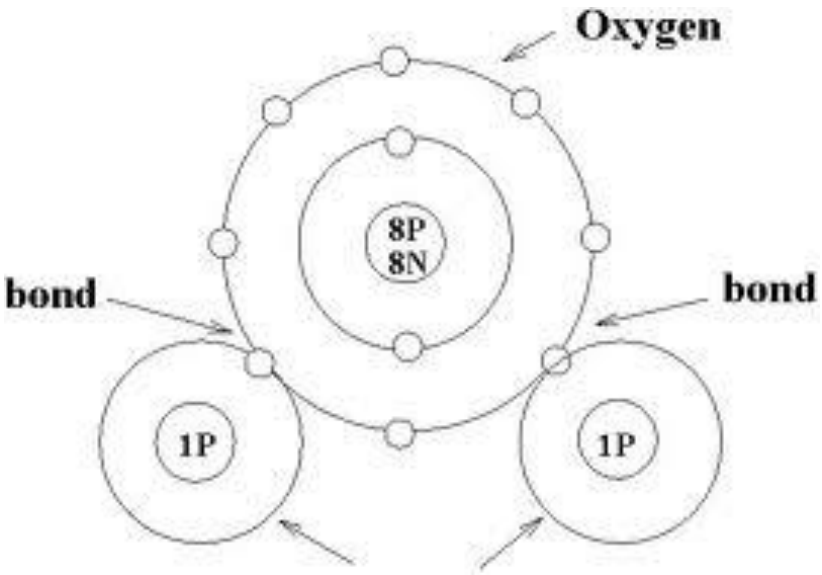
This scales with mass number, A:

$$\underbrace{N_A \times A \times 1.66 \times 10^{-24}}_{\text{`Atomic mass'}} = A \times 1\,\text{gram}$$

(mass of 1 mole of an element)

So `A' grams of every element contains Avogadros' number of atoms. Which leads to the concept of atomic mass.

Molecular mass



The concept of atomic mass extends to molecules too.

Sum of mass nos, means add up the atomic masses of all atoms in molecule

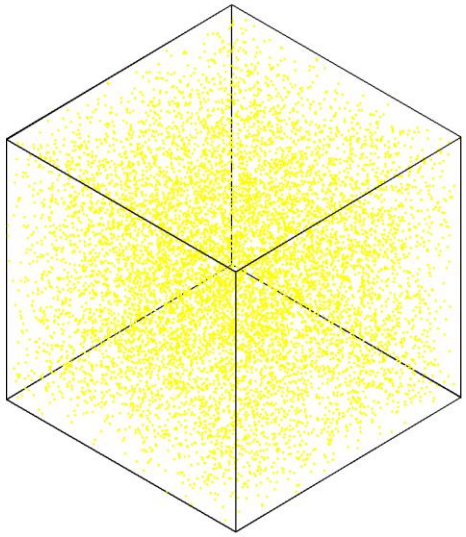
$$N_A \times \text{sum of mass no}^{s} \times 1.66 \times 10^{-24} = \text{molecular mass}\,[\text{g mole}^{-1}]$$

`Molecular mass' (mass of 1 mole of an substance)

So `sum of mass no' grams of every molecule contains Avogadros' number of molecules.

Avogadro's number
(in 'related problems 2')

Know how many atoms / molecules there are per unit mass of a substance (i.e. 1/molecular mass)



How can we work out the number of atoms / molecules per unit volume?



?

$$mass\,[\mathrm{kg}] = volume\,[\mathrm{m}^3] \times density\,[\mathrm{kg\,m}^3]$$

$$m = V \times \rho$$

If you know how many atoms (or molecules) there are per unit volume [$\#m^{-3}$], how can you calculate the *average* distance between atoms (or molecules) [m]?

Aside on Volume/Density

Make sure you know how to calculate the area, volume and surface area of simple shapes e.g. square/circle/triangle and cube/cuboid, sphere and cylinder, cone.

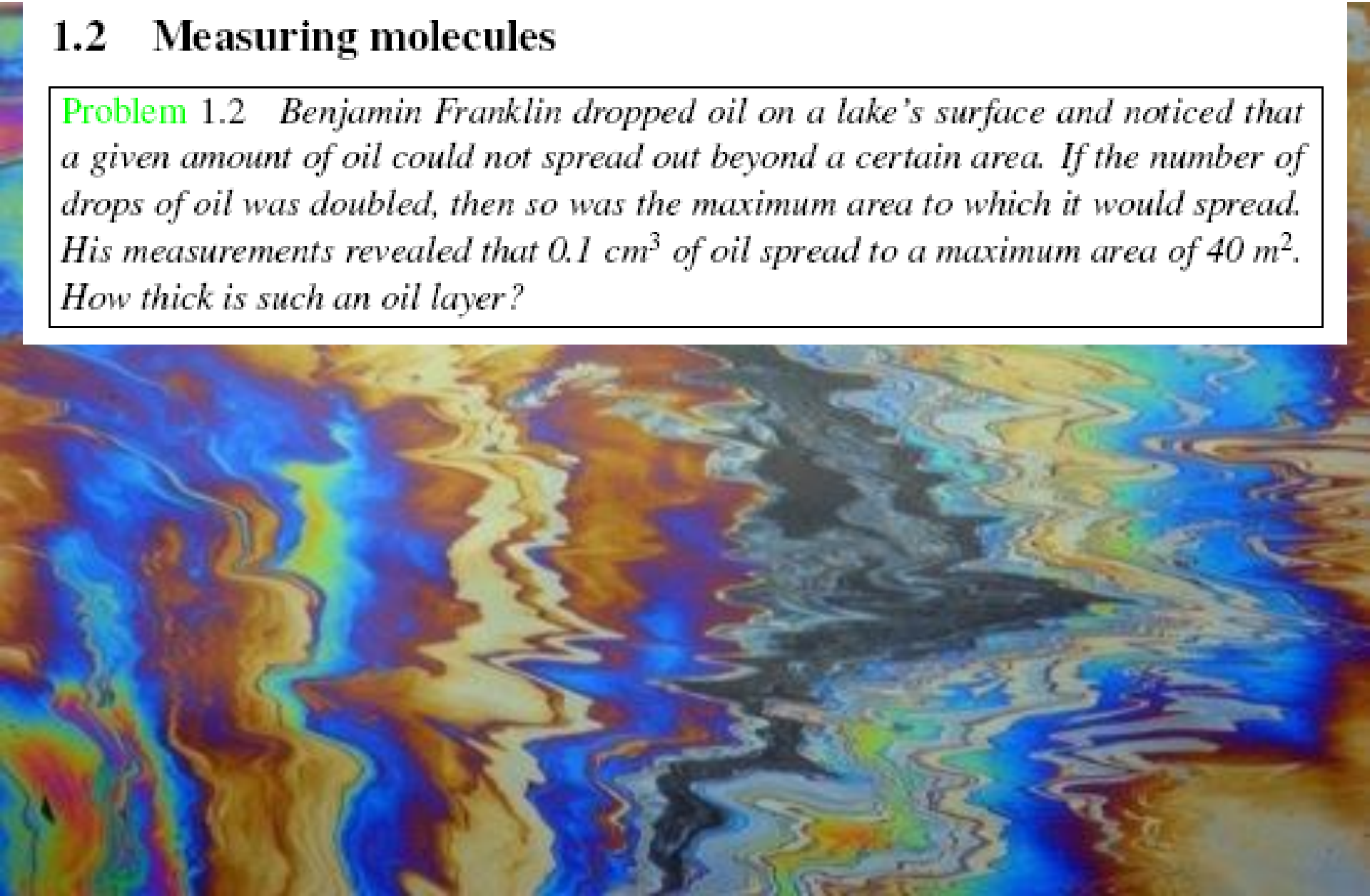
You can find the formulae for these in appendix B of file eart11200_appendices on Blackboard.

You can look up the density of substances, but it will (usually) be stated for STP – beware, density changes with both temperature and pressure!

Problem: How thick is a layer of oil?

1.2 Measuring molecules

Problem 1.2 *Benjamin Franklin dropped oil on a lake's surface and noticed that a given amount of oil could not spread out beyond a certain area. If the number of drops of oil was doubled, then so was the maximum area to which it would spread. His measurements revealed that 0.1 cm^3 of oil spread to a maximum area of 40 m^2. How thick is such an oil layer?*



Today / in week

- First do supplementary material
  - eart11200_additional_1
  - eart11200_additional_2
- Then attempt main problem
  - eart11200_week01
- Then related questions (in same file)

- In the week do the Blackboard assessments as mentioned at the end of the main problem file.

- Next: see you in the practical class (Friday,11am)



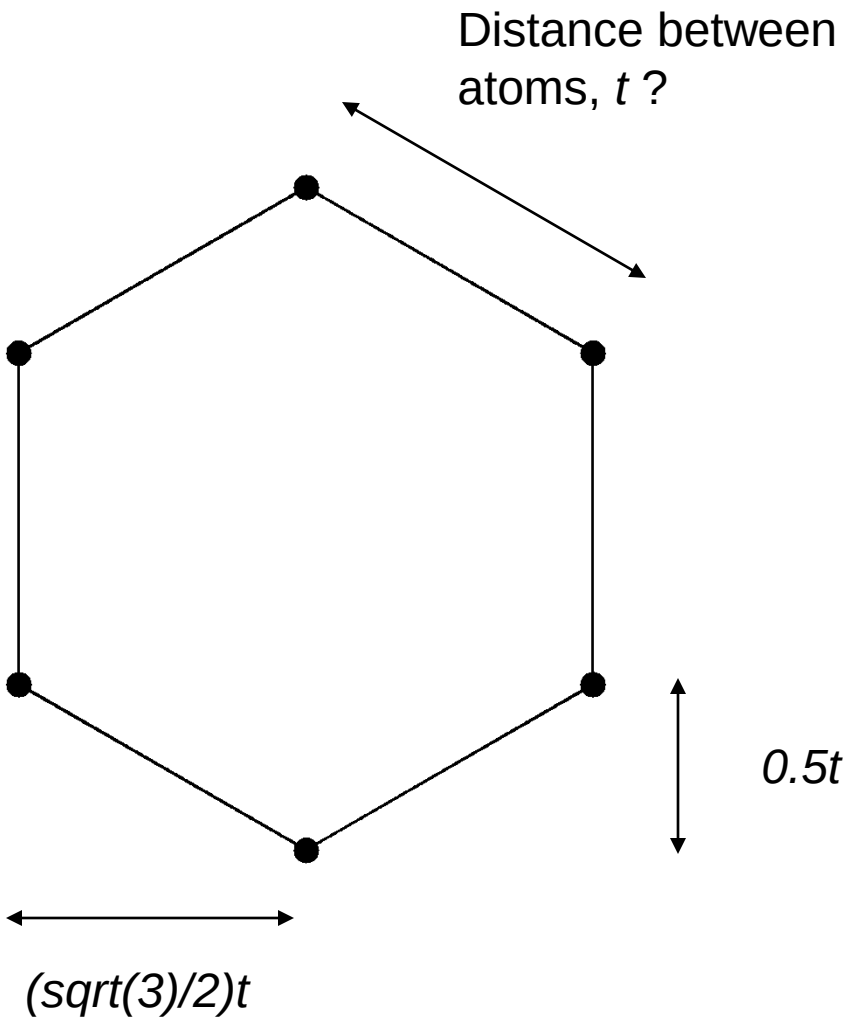




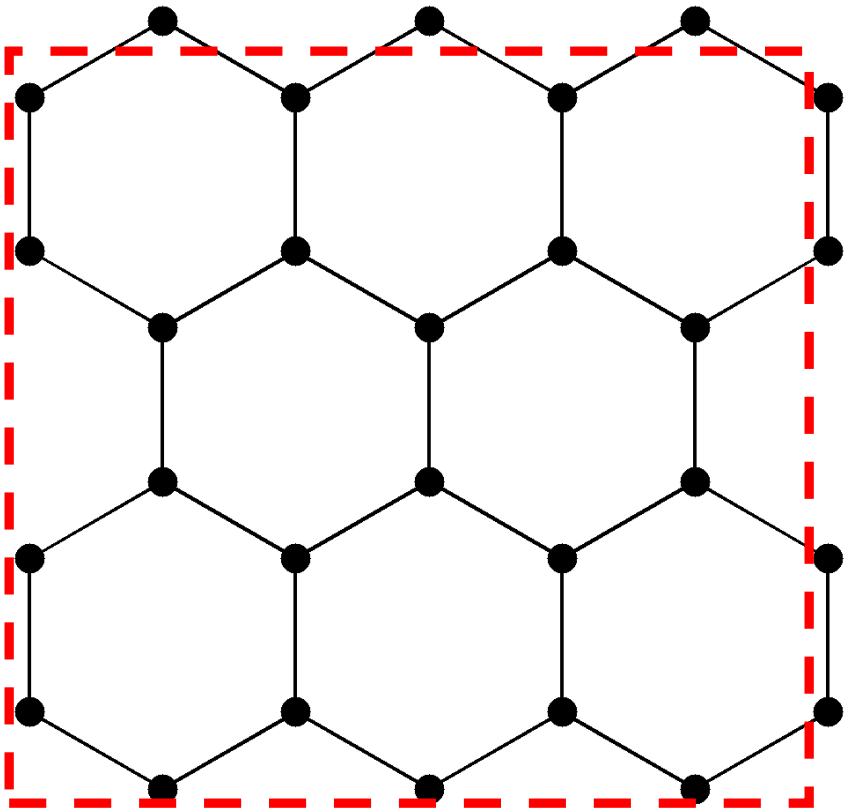
- **Laptop**
- **Pencil and paper**
- **Calculator**

Remember to have everything you need available

The `Graphene’ question



The `Graphene' question



Graphene is 2-D with specific surface area of 1.299 $m^2\ mg^{-1}$ and carbon is 12 $g\ mol^{-1}$

What is the mass of one carbon atom? (hint: Avogadros' number)

Count the atoms within the red-outlined area

Calculate the area of the red 'square' in terms of the bond length, *t*

Take a ratio and set equal to the measured specific surface area.

What is the distance between atoms, *t*?